

Nama : Achmad Pradita Dwi Firmansyah

NIM : 254107020130

Kelas/Absen : TI – 1G / 01

JOBSHEET

Studi Kasus 2

1. Kode Program

Class Pembeli

```
package JobsheetCM2;
```

```
public class Pembeli {  
    int noAntrean;  
    String namaPembeli;  
    String NoHp;  
  
    Pembeli(int a, String b, String c) {  
        noAntrean = a;  
        namaPembeli = b;  
        NoHp = c;  
    }  
  
    void tampil() {  
        System.out.printf("%-15d %-15s %-12s %n", noAntrean, namaPembeli, NoHp);  
    }  
}
```

Class Pesanan

```
package JobsheetCM2;
```

```
public class Pesanan {  
    int kodePesanan;  
    String namaPesanan;  
    int harga;  
  
    Pesanan(int kodePesanan, String namaPesanan, int harga) {  
        this.kodePesanan = kodePesanan;  
        this.namaPesanan = namaPesanan;  
        this.harga = harga;  
    }  
  
    void tampil() {  
        System.out.printf("%-15d %-15s %-12d %n", kodePesanan, namaPesanan, harga);  
    }  
}
```

Class NodePembeli

```
package JobsheetCM2;
```

```
public class NodePembeli {  
    Pembeli dataPembeli;
```

```

NodePembeli prev;
NodePembeli next;

NodePembeli (Pembeli dataPembeli) {
    this.dataPembeli = dataPembeli;
    this.prev = null;
    this.next = null;
}
}

```

Class NodePesanan

package JobsheetCM2;

```

public class NodePesanan {
    Pesanan dataPesanan;
    NodePesanan prev;
    NodePesanan next;

    NodePesanan (Pesanan dataPesanan) {
        this.dataPesanan = dataPesanan;
        this.prev = null;
        this.next = null;
    }
}

```

Class DLLPembeli

package JobsheetCM2;

```

public class DLLPembeli {
    NodePembeli head;
    NodePembeli tail;

    DLLPembeli() {
        head = null;
        tail = null;
    }

    boolean isEmpty() {
        return head == null;
    }

    void tambahAntrian(int a, String b, String c) {
        Pembeli dataPembeli = new Pembeli(a, b, c);
        NodePembeli dataPembeliBaru = new NodePembeli(dataPembeli);
        if (isEmpty()) {
            head = tail = dataPembeliBaru;
        } else {
            tail.next = dataPembeliBaru;
            dataPembeliBaru.prev = tail;
            tail = dataPembeliBaru;
        }
    }

    void cetakAntrian() {

```

```

    if (isEmpty()) {
        System.out.println("Antrian masih kosong.");
        return;
    }

    NodePembeli current = head;
    System.out.println("=====");
    System.out.println("Daftar Antrian Pembeli");
    System.out.println("=====");
    System.out.printf("%-15s %-15s %-12s %n", "No Antrian", "Nama", "No HP");
    while (current != null) {
        current.dataPembeli.tampil();
        current = current.next;
    }
}

void panggilAntrian() {
    if (isEmpty()) {
        System.out.println("Antrian masih kosong.");
        return;
    } else if (head == tail) {
        head = tail = null;
    } else {
        head = head.next;
        head.prev = null;
    }
}
}

```

Class DLLPesanan

package JobsheetCM2;

```

public class DllPesanan {
    NodePesanan head;
    NodePesanan tail;

    DllPesanan() {
        head = null;
        tail = null;
    }

    boolean isEmpty() {
        return head == null;
    }

    void simpanPesanan(int kode, String namaPesanan, int harga) {
        Pesanan pesananBaru = new Pesanan(kode, namaPesanan, harga);
        NodePesanan dataPesananBaru = new NodePesanan(pesananBaru);
        if (isEmpty()) {
            head = tail = dataPesananBaru;
        } else {
            tail.next = dataPesananBaru;
            dataPesananBaru.prev = tail;
            tail = dataPesananBaru;
        }
    }
}

```

```

    }
}

void sortingNamaPesanan() {
    if (isEmpty() || head.next == null) {
        System.out.println("Data pesanan masih kosong.");
        return;
    }
    NodePesanan current = head;
    while (current != null) {
        NodePesanan next = current.next;
        while (next != null) {
            if (current.dataPesanan.namaPesanan.compareTo(next.dataPesanan.namaPesanan) > 0) {
                Pesanan sorted = current.dataPesanan;
                current.dataPesanan = next.dataPesanan;
                next.dataPesanan = sorted;
            }
            next = next.next;
        }
        current = current.next;
    }
}

void cetakPesanan() {
    if (isEmpty()) {
        System.out.println("Data pesanan masih kosong.");
        return;
    }
    sortingNamaPesanan();
    NodePesanan current = head;
    System.out.println("=====");
    System.out.println("LAPORAN PESANAN (URUT NAMA PESANAN)");
    System.out.println("=====");
    System.out.printf("%-15s %-15s %-12s %n", "Kode Pesanan", "Nama Pesanan", "Harga");
    while (current != null) {
        current.dataPesanan.tampil();
        current = current.next;
    }
    System.out.println("=====");
    System.out.println("Total Pendapatan Restoran = Rp." + hitungTotalPendapatan());
}

int hitungTotalPendapatan() {
    int total = 0;
    NodePesanan current = head;
    while (current != null) {
        total += current.dataPesanan.harga;
        current = current.next;
    }
    return total;
}
}

```

Class RestoMain

```
package JobsheetCM2;
```

```

import java.util.Scanner;

public class RestoMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        DLLPembeli antreanPembeli = new DLLPembeli();
        DllPesanan dataPesanan = new DllPesanan();

        antreanPembeli.tambahAntrian(1, "Ainra", "08224500000");
        antreanPembeli.tambahAntrian(2, "Danra", "08224511111");
        antreanPembeli.tambahAntrian(3, "Sanri", "08224522222");
        int a = 4;
        int pilihMenu;
        do {
            System.out.println("=====");
            System.out.println("SISTEM ANTRIAN ROYAL DELISH");
            System.out.println("=====");
            System.out.println("1. Tambah Antrian");
            System.out.println("2. Cetak Antrian");
            System.out.println("3. Hapus Antrian dan Pesan");
            System.out.println("4. Laporan Pesanan");
            System.out.println("0. Keluar");
            System.out.print("Pilih menu : ");
            pilihMenu = sc.nextInt();
            sc.nextLine();
            switch (pilihMenu) {
                case 1:
                    System.out.print("Nama Pembeli : ");
                    String b = sc.nextLine();
                    System.out.print("No HP      : ");
                    String c = sc.nextLine();
                    antreanPembeli.tambahAntrian(a, b, c);
                    System.out.println("Antrian berhasil ditambahkan dengan nomor: " + a);
                    a++;
                    break;
                case 2:
                    antreanPembeli.cetakAntrian();
                    break;
                case 3:
                    if (antreanPembeli.isEmpty()) {
                        System.out.println("Antrian masih kosong.");
                        break;
                    }
                    String namaPembeli = antreanPembeli.head.dataPembeli.namaPembeli;
                    antreanPembeli.panggilAntrian();
                    System.out.print("Kode Pesanan : ");
                    int kode = sc.nextInt();
                    sc.nextLine();
                    System.out.print("Nama Pesanan : ");
                    String namaPesanan = sc.nextLine();
                    System.out.print("Harga      : ");
                    int harga = sc.nextInt();
                    sc.nextLine();

```

```

        dataPesanan.simpanPesanan(kode, namaPesanan, harga);
        System.out.println(namaPembeli + " telah memesan " + namaPesanan);
        break;
    case 4:
        dataPesanan.cetakPesanan();
        break;
    case 0:
        System.out.println("Terima Kasih");
        break;
    default:
        System.out.println("Menu yang dipilih tidak valid.");
        break;
    }
} while (pilihMenu != 0);
}
}

```

2. Hasil Run Terminal

```

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 1
Nama Pembeli : Vania
No HP      : 08422234556
Antrian berhasil ditambahkan dengan nomor: 4
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 2
=====
Daftar Antrian Pembeli
=====
No Antrian    Nama      No HP
1             Ainra     08224500000
2             Danra     08224511111
3             Sanri     08224522222
4             Vania     08422234556

```

```

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 3
Kode Pesanan : 123
Nama Pesanan : Es Teler
Harga      : 5000
Ainra telah memesan Es Teler
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 2
=====
Daftar Antrian Pembeli
=====
No Antrian   Nama           No HP
2            Danra          08224511111
3            Sanri          08224522222
4            Vania          08422234556
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 4
=====
LAPORAN PESANAN (URUT NAMA PESANAN)
=====
Kode Pesanan  Nama Pesanan  Harga
125           Es Degan     6000
123           Es Teler     5000
124           Mie Goreng   34000
=====
Total Pendapatan Restoran = Rp.45000
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 0
Terima Kasih
PS C:\PENYIMPANAN\Documents\G\PraktikumAlgoritmaDanStrukturData>

```